



University of Engineering and Technology, Taxila
Software Engineering Department (2024)

OPP Lab Project Report

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Bookstore Application

1. Introduction

This project involves the creation of a simple online bookstore application using JavaFX, which serves as a platform for users to browse books, make purchases, leave reviews, and manage their account through login and registration features. The application contains various scenes such as a landing page, login page, catalog page, and a review page.

The primary aim of this project is to demonstrate proficiency in JavaFX UI design, handling user events, data validation, and navigation between different application pages. It also emphasizes backend file operations such as user account management (login, signup) and review submissions.

2. Architecture and Components

The project is divided into multiple classes:

LandingPage: This is the main entry point of the application. It offers navigation to other pages like the login page, book catalog, reviews, and shopping cart. The UI is designed with a background image and styled components for better user interaction.

LoginPage: This page allows users to either log in or sign up for an account. It features text fields for username, password, and optionally email (for sign-ups).

CatalogPage: Although not fully implemented in the code shared, this page serves as the catalog of books where users can browse available books.

ReviewPage: This page allows users to submit reviews or feedback about books they've purchased or interacted with.

UserUtil: A utility class that handles user authentication and account creation, which interacts with a file system to store user data (username, password, email).

Main: The main class demonstrates JavaFX event handling using mouse clicks and keyboard presses for interaction tracking.

Key Technologies Used:

JavaFX: For building the graphical user interface (GUI).

BufferedReader and BufferedWriter: For handling file operations such as reading and writing user data.

CSS Styling: For customizing the appearance of UI elements, like buttons and text fields.

3. Key Functionalities and Code Breakdown

3.1 Landing Page

The landing page (LandingPage.java) includes the main interface for navigating the bookstore. It features a navigation bar at the top with buttons for different sections of the application. Each button transitions the user to another scene, such as login, catalog, or review pages. Here's a key snippet of the landing page:

```
HBox navBar = new HBox(20);
navBar.setAlignment(Pos.CENTER_RIGHT);
navBar.setPadding(new Insets(20));

Button btnUser = new Button("👤 User Login/Sign up");
Button btnCatalog = new Button("📖 Book Catalog");
Button btnReview = new Button("★ Reviews");
Button btnCart = new Button("🛒 Cart");

Button[] buttons = {btnUser, btnCatalog, btnReview, btnCart};
for (Button btn : buttons) {
    btn.setStyle("-fx-background-color: #013220; -fx-text-fill: white; -fx-font-weight:
}

btnUser.setOnAction(e -> stage.setScene(new LoginPage().getScene(stage)));
btnCatalog.setOnAction(e -> stage.setScene(new Catalog().getScene(stage)));
btnReview.setOnAction(e -> stage.setScene(new ReviewPage().getScene(stage)));
btnCart.setOnAction(e -> stage.setScene(new CartPage().getScene(stage)));
```

HBox layout is used for the navigation bar, and various buttons are styled for a consistent look.

3.2 Login Page

The login page (LoginPage.java) offers both login and sign-up functionality. It dynamically adjusts the form based on whether the user is in login mode or sign-up mode. Here's an important part of the code for toggling between login and signup modes:

```

Hyperlink switchLink = new Hyperlink("Don't have an account? Sign up");
switchLink.setTextFill(Color.web("white"));
switchLink.setOnAction(e -> {
    isLoginMode = !isLoginMode;
    if (isLoginMode) {
        subheading.setText("Login to continue");
        actionBtn.setText("Login");
        switchLink.setText("Don't have an account? Sign up");
        emailBox.setVisible(false);
    } else {
        subheading.setText("Create a new account");
        actionBtn.setText("Sign Up");
        switchLink.setText("Already have an account? Login");
        emailBox.setVisible(true);
    }
    usernameField.clear();
    passwordField.clear();
    emailField.clear();
    message.setText("");
});

```

This segment switches the UI between the login and signup modes by toggling visibility and updating the button text accordingly.

3.3 Review Page

The review page (ReviewPage.java) allows users to submit a review for books. A TextArea component is used for the review input, and a button submits the form:

```

submitButton.setOnAction(e -> {
    if (usernameField.getText().isEmpty() || reviewField.getText().isEmpty()) {
        showError("Please fill in all fields!");
    } else {
        thankYouMessage.setText("Thank you for your review!");
        thankYouMessage.setVisible(true);
        usernameField.clear();
        reviewField.clear();
    }
});

```

This segment handles the submission of a review and displays a confirmation message.

3.4 User Authentication Utility

The UserUtil.java class is responsible for checking if a user already exists, validating login credentials, and creating new users by writing to a text file. Here's a snippet of the code for validating user login:

```

public static boolean validateLogin(String username, String password) {
    try (BufferedReader reader = new BufferedReader(new FileReader(FILE_NAME))) {
        String line;
        while ((line = reader.readLine()) != null) {
            String[] parts = line.split(",", 3);
            if (parts.length >= 2 && parts[0].equalsIgnoreCase(username) && parts[1].equals(password)) {
                return true;
            }
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
    return false;
}

```

This method reads the user data from a file, validates the username and password, and returns a success or failure response.

4. Conclusion

This project provides an overview of how to create a user-friendly graphical user interface using JavaFX. It integrates navigation between multiple pages, provides form validation, manages user data with file operations, and demonstrates event handling for a smooth user experience. Future enhancements could include integrating a database for user management, adding more features to the book catalog, and implementing a shopping cart for book purchases.

Contributor 1: Cart System & Data Core

Key Components:

1. Cart Page
2. Cart Manager
3. BookItem

Contributor 2: Authentication & Reviews

Key Components:

1. Login Page
2. UserUtil
3. Review Page

Contributor 3: Catalog & Interface

Key Components:

1. Catalog Page
2. Landing Page